Building tutorial for non-geeks

.:[absolute beginners :-)]:.

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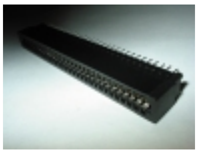
You've got **DivIDE DIY Kit** (Do It Yourself). In the package there is DivIDE mainboard, chip sockets, chips, resistors, one diode, three LED diodes, IDE connector, NMI button, bus connector, ceramic condensators, three transistors, etc... What to do with that stuff? **Let's build your own DivIDE!** It's easier than it seems and even amateur could do it with a bit of patience.

What will you need:

- solder
- soldering iron with resin
- pincers or scissors
- tweezers or small screwdriver, small knife or something similar
- small nail
- saw, nail-file, emery paper (only to smooth the bus connector)
- a small vice could be very helpful, but it isn't necessary

There is such a connector in the package. The ZX Spectrum has **28-pin** long bus (incl. key), so we must modify the connector because it is longer than we want.



01

At first, count all pins (it could vary from piece to piece). You have to have 28 free pins, which means that all the others will go away. Count 28 pins from the middle in such way, that on each end should leave at least 1 free pin. On my one, there was 1 and on the other end 2 pins left. Remove those overlapping pins with tweezers or small screwdriver. You have to remove pins from both sides, up and down of course. The result should look like in the picture.

02

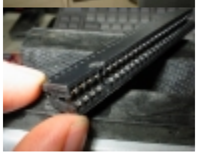
Now, you need do cut the connector from both ends. Here the vice could be helpful. Fix the connector into the vice in such way, that it could be sawed easily and cut off both sides of the connectors (those which you've removed metal pins before).


03

Clean up both sides of the connector with the nail-file and emery paper.

04

Now, it is needed to add a key into the connector. Our small nail will serve as the key. It is necessary, because the interface or the Spectrum could be damaged by wrong insertion of DivIDE to the Spectrum bus.
Place the connector with pins facing up (from you) and count 4 pins from the LEFT. Remove the FIFTH pin in the same way, like before sawing. Here will be the key-nail. As a help you can look at the another ZX Spectrum interface, just like a Kempston joystick interface, etc.
Make a small mark from the top, e.g. using a felt-tip or make a small socket with screwdriver, cca 2 mm far from the edge of the connector. Fix the connector into the vice comfortably. Take the nail using the tongs and place it upright to the mark you've done before. Take a solder to the other hand and hot the nail as long as it melts into the connector. Hold the nail upright to the connector all the time!
You can let the head of the hot nail go into the connector's top.
Chop the nail on the other side and clean up by a nail-file to the level of the connector's bottom.

05

Clean up the connector thoroughly.
Bend both rows of pins in such way, that they touched the pins of the DivIDE mainboard.

06

Make sure, that you have placed the connector to the mainboard in the right way and your key (nail) is inserted on the right place. Solder the first pin of the connector to DivIDE mainboard (doesn't matter which one, but it must be the first on the edge).

07

Arrange the connector straight to the mainboard and solder the last pin on the other side. Then you can just fix it a little.

08

Solder all the pins step by step, on both sides. It's easy and you will need only a small amount of soldering iron.

09

Now, put all the sockets to their right places and beware of the cut-off directions (a small trough on the socket's side). When you look at the DivIDE from the upside, they ALL must face up to the UP or to the RIGHT!
Solder all sockets. Make sure all the them are maximally pressed to the board. You can't move with them after soldering! It applies for all the components (excluding chips).

10

Solder the other not-alive components - pins for jumpers, button and IDE connector. The IDE connector has to face its carve into the DivIDE, like in the picture.

11

Place all the components onto the mainboard. You don't need a scheme, just look at the pictures and make it carefully by them. Just don't place the integrated circuits (chips). Beware of placing LED diodes. The shortest leg of each one has to be closer to the IDE connector! Also be careful by placing the other components. If you don't know the resistor values, orientate by the color stripes and compare with pictures. Don't place any of the component upside down! Only ceramic condensators (here yellow) can be placed as you want.




12

You have a forest of long legs down on the board. Solder them step by step and then cut by pincers or scissors. You can also cut them at first and then solder, but personally I like the first way more. Make sure the components are placed though on the board.


13

Put all the chips into its sockets. Beware of hole orientations again! Pre-programmed GAL chips can be recognized by this key:
Place them upside down (pins heading up). On each GAL are two small circles, one with producer's name and the other, important for us, with a couple of numbers. Sort them from the lowest to the highest number and you will get **A-R-M**. By other words, **GAL_A** (lowest number), **GAL_R** (middle number) and **GAL_M** (highest number). GAL_A is placed under the NMI button, with hole facing right. GAL_M is on the bottom left (hole up) and GAL_R is next to GAL_M on the right (hole up). Insert the other chips looking to the scheme or photos of already built interface.

14

Your DivIDE is done! Just prepare one jumper (it should be a part of the kit, if it's not, take it from some old PC). Put the DivIDE to the Speccy and turn it on. Don't insert a jumper.
The Spectrum should boot normally. If the jumper 'E' is inserted, you will get attribute chaos after switching on. Load up the firmware from tape or from PC (e.g. FatWare, Demfir, MDOS3, etc.). Make sure the jumper 'E' is not inserted (You will need also jumper 'A' inserted only if you have ZX Spectrum +2A(B) or +3) and press any key. The BORDER will flash for a while and then it shows up a message, that everything is ok. **INSERT JUMPER 'E' NOW**. Switch off the Spectrum.

*Firmware is usually distributed as a TAP file. If you want to "beep" it to the Spectrum, you can use TAP2TZX from zxspectrum-utils package, which converts TAP to TZX format and then PLAYTZX, which "beeps" it directly to the soundblaster output or using /voc switch generates a .VOC file, which can be easily played in any common media player. Of course, this is only one of the possible solutions.

15

Put a CF-IDE with Compact Flash card (filled already with some ZX stuff) to DivIDE or connect a 40-pin ribbon cable with hard disk (this will need an extra power supply, e.g. from a PC).

*CF-IDE is not included in DivIDE DIY Kit. You'll have to buy or make it on your own. Some of CF-IDEs had been made by Zilog, then appeared more compact version by Noby. Similar adaptor can be bought in your local PC store as a normal IDE<>CF interface (Sadly, this usually needs power line modification). Those by Zilog and Noby are powered right from the ZX Spectrum bus.

16 Turn on the Spectrum. It should boot up with DivIDE logo. When the Sinclair copyright message appears, press NMI button.

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